

Sujal Dhakal

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EDUCATION

Georgia Institute of Technology

Bachelor's of Science in Computer Science

Expected Graduation, Dec 2026

Atlanta, GA

- **GPA:** 3.9
- **Related Coursework:** Data Structures & Algorithms, Databases & SQL, Software Design & UML, Object-Oriented Programming in Java, Computing in Python, Discrete Mathematics, Linear Algebra

EXPERIENCE

Bits of Good

Jan. 2025 – Present

Software Developer

- Refactored the backend from raw MongoDB to Mongoose, **restructuring 10 micro-services and 23 API endpoints**, enhancing maintainability and type safety.
- Built a multi-screen medication management system in NextJS and TailwindCSS, **creating 17 reusable components** and integrating Zod for user validation, **reducing input errors by 60%**.
- Designed a secure password reset system with NodeJS, Mongoose, and BoG's internal email tool, streamlining account recovery for users.

Technique Newspaper

Nov. 2024 – Present

Software Developer

[Website](#)

- Rebuilt Georgia Tech's student newspaper website with NextJS and TailwindCSS, replacing WordPress's builder while maintaining WordPress as a CMS. **Decreased load times by 8000ms and improved SEO rankings.**
- Engineered a custom service wrapper for the WordPress REST API, **reducing API call redundancy by 55%** through caching and simplifying access to posts, images, authors, and categories.
- Constructed a modular front-end with reusable story components and an optimized search feature.

Personality Academy

Aug. 2020 – Aug. 2024

Co-Founder

[Website](#)

- Launched a personality education platform **attracting 3,000+ registered users** using NextJS and Supabase with courses, educative tools, and Stripe integration for donations.
- Implemented a **custom OAuth system** with Google, Discord, and email validation, utilizing Prisma ORM.
- Developed the TypeSearch tool for **querying 2,000+ typed individuals**, featuring **multi-filter search**, and **dynamic diagrams for 512 unique types** with images managed by ImageKit.
- Created the TypeChart tool, allowing users to explore personality types interactively and generating unique behavioral spectrums from a 10-digit type code.

PROJECTS

D.A.N. | TypeScript, Python, React Native, NodeJS, Firebase, TensorFlow, Whoop, Spotify, Expo, Git

[GitHub](#)

- Implemented dementia management into a mobile app, **utilizing Whoop OAuth**, with real-time health data updates via **Firebase Cloud Functions and Webhooks**.
- Built an agitation detection algorithm through the Whoop API, analyzing 5 health metrics to trigger music with the Spotify API or alerts for **95% accuracy in agitation detection**.
- Developed a wander tracking model with TensorFlow, **detecting wandering behavior with 90% accuracy**.

Check-In | JavaScript, NextJS, TailwindCSS, PostgreSQL, Neon, Prisma, Vercel, Git

[Website](#) | [GitHub](#)

- Created an automated attendance management system, handling **1,000+ students** across **15+ organizations**, reducing event start-up time by over 40 minutes.
- Designed a dynamic dashboard for tracking attendance and events, offering **real-time analytics** and **customizable parameters** (e.g., recurring events, tardy tracking).
- Incorporated CSV import/export for seamless data management, resulting in **100% organization adoption** within the first month of deployment in a high school district.

TECHNICAL SKILLS

Languages: Python, HTML, CSS, JavaScript, TypeScript, PHP, SQL, Java, C#, C++

Frameworks: Flask, Django, React, React-Native, NodeJS, NextJS, TensorFlow, Pandas, NumPy, Matplotlib

Technologies: Git, Docker, Socket.IO, Google Cloud, Firebase, Vercel, Netlify, Heroku, Stripe, Postman, MongoDB